

ZION MISSION SCHOOL

COMPUTER SCIENCE PRACTICAL – CLASS XII

JAVA PROGRAMMING

Programming Language: Java

Software: Visual Studio Code

Compiler/Runtime: Java JDK

Practical 1 – Display User Information

Question:

Write a Java program to accept the **name and age of a user** from the keyboard and display the entered name and age.

Concepts Used: Scanner, variables, String, int, input and output.

Practical 2 – Addition of Two Numbers

Question:

Write a Java program to accept **two integer numbers** from the user and calculate and display their **sum**.

Concepts Used: Scanner, integer variables, arithmetic operator (+) and output.

Practical 3 – Voting Eligibility

Question:

Write a Java program to accept the **age of a person** and determine whether the person is **eligible or not eligible to vote**. A person is eligible to vote if their age is **18 years or above**.

Concepts Used: Scanner, if-else statement, relational operator ($\geq$).

Practical 4 – Student Grade

Question:

Write a Java program to accept the **marks obtained by a student** and display the appropriate grade according to the following conditions:

Marks	Result
90–99	Grade A
75–89	Grade B
60–74	Grade C
Below 60	Fail
0 or 100 and above	Invalid Marks

Concepts Used: Scanner, if-else-if ladder, relational and logical operators.

Practical 5 – Voting Eligibility Using Nested If

Question:

Write a Java program to check whether a person is eligible to vote based on **age and citizenship**.

The program should accept:

- The person's **age**.
- Their **citizenship status** as true or false.

A person can vote only when:

- The age is **18 years or above**, and
- The person is a **citizen**.

Display appropriate messages for the different conditions.

Concepts Used: Scanner, Boolean data type, if-else, **nested if**, relational and conditional statements.

Practical 6 – Café Menu Using Switch Case

Question:

Write a Java program to create a simple **Café Menu** using the switch-case statement.

Display the following menu:

Choice	Item	Price
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1	Burger	\$5.99
2	Pizza	\$8.99
3	Pasta	\$7.50
4	Soda	\$1.99

Accept the customer's choice from the keyboard and display:

- The selected food item.
- Its price.
- The total bill.
- A suitable thank-you message.

If the user enters a choice other than **1–4**, display an **Invalid Choice** message.

Concepts Used: Scanner, switch-case, case, break, default, variables, conditional statements and formatted output.

Practical Copy Format

For **each practical**, students should maintain the following format in their practical copy:

Ruled Page:

1. Practical No.
2. Date
3. Aim / Question
4. Tools Required
5. Algorithm / Steps
6. Java Program
7. Result

Blank/White Page:

- Paste the **printed output/screenshot of the program execution** neatly using glue.

Note: All programs must be written, compiled and executed in the **school computer lab** using **Visual Studio Code** and **Java JDK**.